A T M E
College of Engineering

Department of CSE-Artificial Intelligence & Machine Learning

Database Management System-BCS403

Lab Program No 4

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Lab Program 4

Create a row level trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table.

This trigger will display the salary difference between the old & new Salary.

CUSTOMERS(ID,NAME,AGE,ADDRESS,SALARY)

Triggers in SQL

- A trigger in SQL is a special type of stored procedure that automatically executes when a specific event occurs in a database.
- Triggers can be used to enforce business rules, maintain audit trails, and synchronize related tables.

Types of Triggers

SQL triggers can be classified based on when they execute and the type of event they respond to.

1. Based on Execution Timing

BEFORE Trigger → Executes before an operation (INSERT, UPDATE, DELETE).

AFTER Trigger → Executes after an operation (INSERT, UPDATE, DELETE).

INSTEAD OF Trigger → Replaces the execution of an operation (commonly used in views).

2. Based on Events

INSERT Trigger → Fires when a new record is inserted.

UPDATE Trigger → Fires when an existing record is updated.

DELETE Trigger → Fires when a record is deleted.

SQL Trigger Syntax

A trigger is created using the CREATE TRIGGER statement.

Example: BEFORE INSERT Trigger

```
CREATE TRIGGER before_insert_employee
BEFORE INSERT ON employees
FOR EACH ROW
BEGIN
  SET NEW.created_at = NOW();
END;
```

This trigger sets the created_at field to the current timestamp before inserting a new employee.

AFTER UPDATE Trigger

```
CREATE TRIGGER after_update_salary
AFTER UPDATE ON employees
FOR EACH ROW
BEGIN
  INSERT INTO salary_changes (emp_id, old_salary, new_salary,
change_date) VALUES (OLD.emp_id, OLD.salary, NEW.salary,
NOW());
END;
```

- This trigger logs salary changes whenever an update occurs in the employees table.

INSTEAD OF DELETE Trigger

```
CREATE TRIGGER instead_of_delete
INSTEAD OF DELETE ON employees_view
FOR EACH ROW
BEGIN
  UPDATE employees SET is_deleted = 1 WHERE id =
OLD.id;
END;
```

- This prevents actual deletion from employees_view and marks the record as deleted instead.

create database CUSTOMERS
USE CUSTOMERS;

- CREATE TABLE CUSTOMERS (
- ID INT PRIMARY KEY,
- NAME VARCHAR(255),
- AGE INT,
- ADDRESS VARCHAR(255),
- SALARY DECIMAL(10, 2));

- INSERT INTO CUSTOMERS VALUES (1, 'Ajay Hegde', 50, 'Mysuru',50000.00);
- INSERT INTO CUSTOMERS VALUES (2, 'Satynaryana Rao ', 65,'Mandya', 45000.00)
- INSERT INTO CUSTOMERS VALUES (3, 'Ravishankar', 35,'Shivmogga', 60000.00)
- INSERT INTO CUSTOMERS VALUES (4, 'Anish Behl', 28,'Manglore', 80000.00)
- INSERT INTO CUSTOMERS VALUES (5, 'Hari das Upadaya', 28,'Udupi', 90000.00)

- SELECT * FROM CUSTOMERS

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- DELIMITER //
- **CREATE TRIGGER** after_insert_salary_difference
- AFTER INSERT ON CUSTOMERS
- FOR EACH ROW
- BEGIN
- SET @my_sal_diff = CONCAT('salary inserted is ', NEW.SALARY);
- END;//
- DELIMITER ;

- INSERT INTO CUSTOMERS VALUES (6, 'amit kumar', 38,'bijapur', 90000.00)
- select @my_sal_diff as SALARY_INSERTED

- DELIMITER //
- CREATE TRIGGER after_update_salary_difference
- AFTER UPDATE ON CUSTOMERS
- FOR EACH ROW
- BEGIN
- DECLARE old_salary DECIMAL(10, 2);
- DECLARE new_salary DECIMAL(10, 2);
- SET old_salary = OLD.SALARY;
- SET new_salary = NEW.SALARY;
- SET @my_sal_diff = CONCAT('salary difference after update is ', NEW.SALARY -
- OLD.SALARY);
- END;//
- DELIMITER ;

OUTPUT:

INSERT INTO CUSTOMERS VALUES (6, 'amit kumar', 38,'bijapur', 90000.00)

UPDATE CUSTOMERS SET SALARY=95000 WHERE ID=6

SELECT @my_sal_diff AS salary_differnce

- -- DELETE TRIGGER
- DELIMITER //
- CREATE TRIGGER after_delete_salary_difference
- AFTER DELETE ON CUSTOMERS
- FOR EACH ROW
- BEGIN
- SET @my_sal_diff = CONCAT('salary deleted is ', OLD.SALARY);
- END;//
- DELIMITER ;

select * from CUSTOMERS

ID	NAME	AGE	ADDRESS	SALARY
1	Ajay Hegde	50	Mysuru	50000.00
2	Satynaryana Rao	65	Mandya	45000.00
3	Ravishankar	35	Shivmogga	60000.00
4	Anish Behl	28	Manglore	80000.00
5	Hari das Upadaya	28	Udupi	90000.00
6	amit kumar	38	bijapur	95000.00
NULL	NULL	NULL	NULL	NULL